Ref. [57] — First-Order Methods Almost Always Avoid Saddle Points: The Case of Vanishing Step-Sizes

Ioannis Panageas, Georgios Piliouras, Xiao Wang | NeurIPS 2019

1. Research question: does saddle avoidance survive a decaying step size?

Earlier stable-manifold arguments established that sufficiently small *constant*-step gradient descent (GD) almost surely does not converge to a strict saddle. In actual optimization, step sizes often vary with the iteration. The paper asks whether the same avoidance statement survives this time dependence, including when the step size vanishes. It addresses GD, mirror descent, manifold gradient descent and proximal-point-type methods, not just one Euclidean iteration.

2. Rebuild the geometry from first principles

For smooth $f : \mathbb{R}^d \to \mathbb{R}$, GD with a varying step size is

$$x_{k+1} = G_k(x_k) = x_k - \alpha_k \nabla f(x_k), \qquad \alpha_k > 0.$$

At a stationary point x^* with $\nabla f(x^*) = 0$, linearize the update by writing $e_k = x_k - x^*$:

$$e_{k+1} \approx \big(I - \alpha_k H\big) e_k, \qquad H = \nabla^2 f(x^*).$$

An eigenvector v with $Hv = -\gamma v$ ($\gamma > 0$) has scalar error $z_k = v^\top e_k$ evolving as

$$z_{k+1} \approx (1 + \gamma\alpha_k) z_k, \qquad z_t \approx z_0 \prod_{k<t} (1 + \gamma\alpha_k).$$

The multiplier is greater than one, so negative curvature locally repels perturbations away from the saddle. This is a *teaching linearization*, not the full global theorem. The remaining issue is whether almost every starting point is repelled when the dynamics change with k.

3. What method and result does the paper actually provide?

The authors establish a stable-manifold theorem for *time-nonhomogeneous* dynamical systems, since ordinary autonomous stable-manifold theorems cannot be applied directly to the sequence G_k. They use it to extend almost-sure avoidance to several first-order methods, including isolated and non-isolated strict-saddle sets, for suitably controlled vanishing step sizes. An important regime has $\alpha_k = \Omega(1/k)$ in addition to the method-specific small-step/regularity requirements.

> **Principal result.** Under its smoothness, step-size and initialization hypotheses, the set of initial conditions whose iterates converge to strict saddle points has measure zero. This is an *asymptotic, almost-everywhere* statement, not a bound on how many iterations an escape requires.

4. Why the step-size condition is meaningful

If $\sum_k \alpha_k = \infty$ and the steps are small, then

$$\log \prod_{k<t} (1 + \gamma\alpha_k) = \sum_{k<t} \log(1 + \gamma\alpha_k) \sim \gamma \sum_{k<t} \alpha_k \to \infty.$$

By contrast, when $\sum_k \alpha_k < \infty$, the product can stay bounded. For a simple quadratic $f(x, y) = \frac{1}{2}x^2 - \frac{1}{2}y^2$, the unstable coordinate obeys $y_{k+1} = (1 + \alpha_k) y_k$: if the total step length is finite, its growth may be limited. This example explains the intuition behind why steps that decrease *too fast* can change the conclusions; it is not a replacement for the paper's nonautonomous manifold proof.

5. Assumptions, limitations, and survey relevance

The main arguments assume appropriate differentiability and controlled step sizes and use generic/random initialization. The result does not cover arbitrary discontinuous gradients at ReLU kinks, does not establish avoidance of fully degenerate saddles (where no negative Hessian eigenvalue exists), and does not provide a polynomial hitting time or global optimum guarantee.

Do not conflate: "almost surely does not converge to a strict saddle" $\neq$ "escapes every saddle quickly" $\neq$ "finds the global minimum."

This belongs to **RQ2** (gradient dynamics), **RQ3** (limits of escape claims), and **RQ4** (smoothness and schedule assumptions).

6. Our notes

This paper extends Lee et al.'s strict-saddle avoidance story to learning rates that decay over time. The intuition is that negative-curvature directions still amplify small errors, but proving it needs a time-varying stable-manifold argument. It is useful for our survey because it makes clear that almost-sure avoidance is a statement about where training can end up, not how quickly it gets there. **Original source:** https://papers.neurips.cc/paper_files/paper/2019/hash/3fb04953d95a94367bb133f862402bce-Abstract.html. **Verification:** conference record and original manuscript; the eigenmode example is pedagogical.